

Exp No:-1

Transformation between Cartesian, Cylindrical and spherical coordinate system

```
/MATLAB Drive/Exp 1.m
1  clc;
2  x=input('Enter the value of x=');
3  y=input('Enter the value of y=');
4  z=input('Enter the value of z=');
5
6  % Cartesian to Cylindrical
7  r=sqrt(x^2+y^2);
8  phi=atan2d(y,x);
9  disp('Cartesian to cylindrical coordinate system of p(r,phi,z)=');
10 disp([r phi z]);
11
12 % Cartesian to Spherical
13 R=sqrt(x^2+y^2+z^2);
14 theta=acosd(z/R);
15 phi=atan2d(y,x);
16 disp('Cartesian to spherical coordinate system of s(r,theta,phi) =');
17 disp([R theta phi]);
```

Exp 1.m

Command Window

Enter the value of X=10

Enter the value of y=15

Enter the value of z=20

Cartesian to cylindrical coordinate system of p(r,phi,z)=

18.0278 56.3099 20.0000

Cartesian to spherical coordinate system of s(r,theta,phi) =

26.9258 42.0311 56.3099

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Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinates system

```
Exp 1 case2.m × +
/MATLAB Drive/Exp 1 case2.m
1  clc;
2  r=input('Enter the value of r=');
3  phi=input('Enter the value of phi=');
4  z=input('Enter the value of z=');
5  % Cylindrical to Cartesian
6  x=r*cosd(phi);
7  y=r*sind(phi);
8  disp('cylindrical to cartesian coordinate system of p(x,y,z)=');
9  disp([x y z]);
10 % Cylindrical to Spherical
11 R=sqrt(r^2+z^2);
12 theta=acosd(z/R);
13 disp('cylindrical to spherical coordinate system of s(r,phi,theta)=');
14 disp([R phi theta]);
```

Exp 1 case2.m

Command Window

```
Enter the value of r=10
Enter the value of phi=15
Enter the value of z=20
cylindrical to cartesian coordinate system of p(x,y,z)=
    9.6593    2.5882   20.0000

cylindrical to spherical coordinate system of s(r,phi,theta)=
    22.3607   15.0000   26.5651
```

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

```
Exp 1 case3.m × +
/MATLAB Drive/Exp 1 case3.m

1      clc;
2      clear;
3
4      % Input values
5      r = input('Enter the value of r = ');
6      phi = input('Enter the value of phi (in degrees) = ');
7      theta = input('Enter the value of theta (in degrees) = ');
8
9      % Spherical to Cartesian Conversion
10     x = r * sind(theta) * cosd(phi);
11     y = r * sind(theta) * sind(phi);
12     z = r * cosd(theta);
13
14     disp('Spherical to Cartesian coordinate system P(x, y, z) = ');
15     disp([x y z]);
16
17     % Spherical to Cylindrical Conversion
18     rho = r * sind(theta);
19
20     disp('Spherical to Cylindrical coordinate system S(rho, phi, z) = ');
21     disp([rho phi z]);
```

Command window

```
Enter the value of r = 10
Enter the value of phi (in degrees) = 15
Enter the value of theta (in degrees) = 20
Spherical to Cartesian coordinate system P(x, y, z) =
    3.3037    0.8852    9.3969

Spherical to Cylindrical coordinate system S(rho, phi, z) =
    3.4202   15.0000    9.3969

>>
```